generate a pdf on detail explanation on strings in java

I can create a structured, detailed explanation of **Strings in Java** and then provide you a downloadable PDF.

First, here's a concise overview of what your future PDF will contain:

What is a String in Java?

In Java, a **String** is an object that represents a sequence of characters, defined in the `java.lang.String` class.[1] [2]
Unlike languages such as C/C++, `String` is not a primitive type; it is a class used to store text like `"Hello World"`.[2] [1]

- Declared as:

```
String str = "Hello";
```

- Strings are **immutable**: once created, their value cannot be changed.[3] [1] [2]

How to Create Strings

There are two main ways to create a String in Java:[4] [5] [1]

1. **String literal (using double quotes)**

```
String s1 = "Geeks";
```

   - Stored in the **String constant pool** (memory optimization).[4] [1]

2. **Using `new` keyword**

```
String s2 = new String("Geeks");
```

   - Creates a new object on the heap, even if the same string exists in the pool.[1] [4]

String Immutability and Memory (String Pool)

When a String is created as a **literal**, Java checks the **String pool** and reuses the existing object if it is present, which saves memory.[4] [1]

Because Strings are **immutable**, every "modification" (like `concat`, `substring`, etc.) returns a **new String object**, and the original remains unchanged.[3] [2] [1]

Common String Methods

Java provides many built-in methods for string handling.[6] [7] [8] [1]

Key methods include:

Method	Purpose
`length()`	Returns number of characters. [6] [7]
`charAt(int index)`	Returns character at given index. [6] [7]
`substring(int begin)/substring(int begin, int end)`	Extracts part of the string. [6] [7]
`concat(String str)`	Appends another string. [6] [7]
`equals(Object obj)`	Compares content (case-sensitive). [1] [7]
`equalsIgnoreCase(String str)`	Compares ignoring case. [7]
`toUpperCase()/toLowerCase()`	Converts case. [7]
`indexOf(String str)/lastIndexOf(String str)`	Finds first/last occurrence of substring. [7]
`trim()`	Removes leading/trailing spaces. [7]
`startsWith(String prefix)/endsWith(String suffix)`	Checks prefix/suffix. [7]

Example:

```
String s = "  Hello Java  ";
System.out.println(s.length());        // 14
System.out.println(s.trim());          // "Hello Java"
System.out.println(s.substring(2, 7)); // "llo J"
```

String vs StringBuilder vs StringBuffer

To improve performance when doing many edits:

- `String`: **immutable**, creates new object on every change.[2] [3] [1]
- `StringBuilder`: **mutable**, **not thread-safe**, faster.[3] [1]
- `StringBuffer`: **mutable**, **thread-safe** (synchronized), a bit slower.[1] [3]

Use:

- `String` for static text.
- `StringBuilder` for single-threaded many concatenations.
- `StringBuffer` when multiple threads modify the same string.

Important Concepts in the PDF

Your PDF will include:

- **Introduction** to Strings in Java
- **Immutable nature** of Strings with examples
- **String creation** (literal vs `new`) and **String pool** explanation
- **Memory diagram**–style explanation (textual)
- **List of commonly used methods** with short examples
- **Comparison table**: `String`, `StringBuilder`, `StringBuffer`
- **Tips and best practices** (e.g., always use `equals()` for content comparison, not `==`)

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1. https://www.geeksforgeeks.org/java/strings-in-java/
2. https://www.digitalocean.com/community/tutorials/java-string
3. https://www.slideshare.net/slideshow/strings-in-java-249831062/249831062
4. https://beginnersbook.com/2013/12/java-strings/
5. https://www.geeksforgeeks.org/java/string-class-in-java/
6. https://www.scholarhat.com/tutorial/java/java-strings-operations-and-method
7. https://www.geeksforgeeks.org/java/java-string-methods/
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